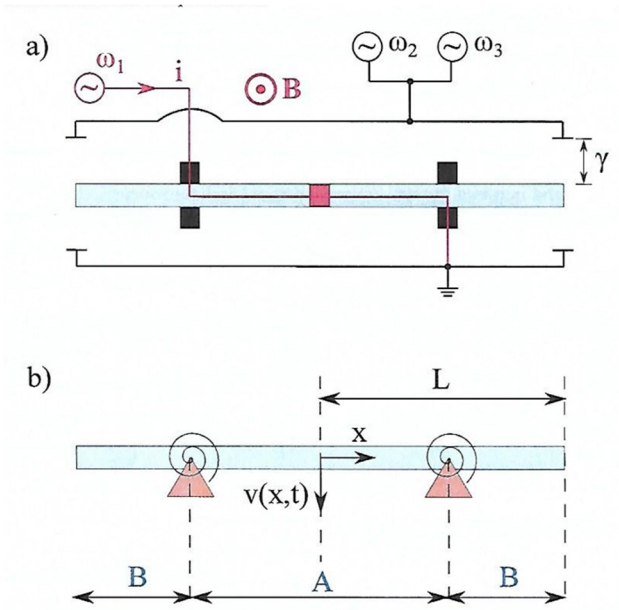
Analytical Dynamics – Project 2 - Appendix – Internal Resonance Nanomechanical Oscillator

Motivated by the investigation of a multiple-span nanomechanical dynamical systems [Alcorta et al 2021] we formulate an initial-boundary-value problem for the beam interaction, employ separation of variables using a Galerkin assumed-mode ansatz to obtain a reduced-order nondimensional model for the nonlinear interaction of the beams operating near their primary resonance.



1. Problem Formulation

The nondimensional equations of motion are:

$$y_{1\tau\tau} + \zeta\, y_{1\tau} + \omega_1^2 y_1 + \alpha_{11} y_1^3 + \alpha_{12} y_1 y_2^2 + \eta_{11} y_1^2 y_{1\tau} + \eta_{12} y_1 y_2 y_{2\tau} = f_1 + g_1$$

$$y_{2\tau\tau} + \zeta\, y_{2\tau} + \omega_2^2 y_2 + \alpha_{21} y_1^2 y_2 + \alpha_{22} y_2^3 + \eta_{11} y_1^2 y_{1\tau} + \eta_{12} y_1 y_2 y_{2\tau} = g_2$$

Where the localized midspan magnetic force (f) and the electrostatic forces at the edges (g) are:

$$f_1 = \mu \sin(\Omega\tau)$$

$$g_{12} = \beta\left(\left[1-(y_1+y_2)\right]^{-2} \pm \left[1-(y_1-y_2)\right]^{-2}\right)$$

Note that all the parameters are positive.

2. Equilibrium Analysis

2.1 Determine the conditions for a stable equilibrium of your unforced (μ=0) and damped autonomous subsystem. Formulate the range of parameters that ensure their stability [$\mathrm{Re}(\lambda_n)<0$, n=1-4].

2.2 Derive the undamped! natural frequencies of the linearized dynamical system about the stable equilibrium [e.g. $\omega_{1,2}$ from $Ay_{\tau\tau} + By = 0$ where $y=(y_1, y_2)^T$ and A, B are constants].

Note possible conditions for an internal resonance [($\omega_2 \sim m*\omega_1$ where m=1,2,3)].

3. Asymptotic Analysis of the Modulated System

3.1 Consider small displacements about the stable equilibrium position and expand the equations of motion above to cubic order where the damping and forcing parameters are sufficiently small.
Rewrite your equations of motion in the form ready singular perturbation.

$$x_{1\tau\tau} + \omega_1^2 x_1 = \varepsilon F_1(x_1, x_{1\tau}, x_2, x_{2\tau}, \tau)$$
$$x_{2\tau\tau} + \omega_2^2 x_2 = \varepsilon F_2(x_1, x_{1\tau}, x_2, x_{2\tau}, \tau)$$

Where the right-hand-sides include: (i) linear damping terms, (ii) forcing term, (iii) quadratic and cubic contributions of the interaction terms.

3.2 Derive the order equations where the first order includes only the linear conservative components. Note your small parameters clearly.

3.3. Assume a periodic multiple scale solution form for the homogeneous first order equations and derive the solvability conditions required to ensure that the asymptotic solution does NOT grow without bound.

3.4 Derive the coupled slowly varying evolution equations from the (complex) solvability condition.

3.5 Determine the non-trivial equilibria of the amplitude evolution equations (which correspond to the amplitudes of the gyro dependent variables) and formulate the conditions of stability of these equilibria.

*Plot the amplitudes as a function of the external forcing frequency (external excitation near primary resonance OR parametric excitation near the principle parametric resonance).

4. Closing Remarks

Summarize your results and conclusions from: (i) equilibrium stability and the conditions for internal resonance (ii) frequency response of the system due to external OR parametric excitation. Comment on possible existence of quasiperiodic or non-stationary response (e.g. there exists a Hopf bifurcation in the slowly varying evolution equations).

5. OPTIONAL BONUS (20%)

Verify your findings numerically using a fourth-order Runge-Kutta ODE solver [e.g. Matlab - ODE45 with sufficiently small tolerances ~ $O(10^{-8})$]: (i) solution of the slowly varying evolution equations vs (ii) solution of the original system.

Comment on accuracy of the (approximate) analytical maximal amplitudes in the frequency response function deduced form the slowly varying evolution equations in comparison to selected maximal amplitudes of the periodic response in the original sleeping gyro equations of motion.